

Quantifying the Internal Validity of Weighted Estimands

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What do we mean by weighted estimands?

Estimating averages of treatment effects is an important objective of empirical research

Applied researchers usually believe treatment effects are heterogeneous

Yet many researchers favor using simple linear models combined with well-established estimators that weren't necessarily designed with heterogeneity in mind, e.g. OLS, IV/2SLS, TWFE

Justifications: computational simplicity, comparability across studies, effectiveness at incorporating high-dimensional covariates, etc.

Other feature: the resulting estimands can often be interpreted as **weighted averages of underlying treatment effects**

What do we mean by weighted estimands? Example: OLS

D = binary treatment variable $\in \{0, 1\}$

$Y(1), Y(0)$ = treated and untreated potential outcomes

$Y = DY(1) + (1 - D)Y(0)$ = realized outcome

X = vector of covariates

$\tau_0(X) = \mathbb{E}[Y(1) - Y(0) \mid X]$ = conditional average treatment effect (CATE)

Suppose unconfoundedness and overlap hold:

$$(Y(1), Y(0)) \perp\!\!\!\perp D \mid X \quad \text{and} \quad \mathbb{P}(D = 1 \mid X) \in (0, 1) \text{ a.s.}$$

Example: OLS

Common practice with covariates: Run a regression of Y on $(1, D, X)$ and report the coefficient on D

In this case and when $p(X) \equiv \mathbb{E}[D \mid X]$ is linear in X , by Angrist (1998)

$$\beta_{\text{OLS}} = \mathbb{E} \left[\frac{\text{var}(D \mid X)}{\mathbb{E}[\text{var}(D \mid X)]} \cdot \tau_0(X) \right],$$

where $\text{var}(D \mid X) = p(X)(1 - p(X))$.

This is a **weighted average of CATEs** where the weights are nonnegative and integrate/sum to 1

This paper

We consider weighted estimands of the kind

$$\mu(a, \tau_0) \equiv \mathbb{E}[a(X)\tau_0(X) \mid W_0 = 1]$$

where

$W_0 \in \{0, 1\}$ is a random variable denoting a subpopulation

$a(X)$ is a weight function that integrates to 1, i.e., $\mathbb{E}[a(X) \mid W_0 = 1] = 1$

$\tau_0(X) = \mathbb{E}[Y(1) - Y(0) \mid W_0 = 1, X]$ is a CATE function

- For OLS, $W_0 = 1$ a.s., $a(X) = \text{var}(D \mid X) / \mathbb{E}[\text{var}(D \mid X)]$, and $\tau_0(X) = \mathbb{E}[Y(1) - Y(0) \mid X]$
- Will see that IV/2SLS with covariates and TWFE with variation in treatment timing also fit this framework

This paper

Two main questions we seek to answer:

1. When does $\mu(a, \tau_0)$ correspond to an average

$$\mathbb{E}[Y(1) - Y(0) \mid W^* = 1]$$

where $W^* \in \{0, 1\}$ is a (possibly latent) subpopulation of W_0 ?

2. What is the largest *size* of this population W^* , or size relative to W_0 , as measured by

$$\mathbb{P}(W^* = 1) \quad \text{and} \quad \mathbb{P}(W^* = 1 \mid W_0 = 1)?$$

The answer to these questions will depend on what is assumed about $\tau_0(X)$.
E.g. is τ_0 completely unrestricted? Assumed to be known?

This paper

When τ_0 is unrestricted, we will show the answers are

1. $\mu(a, \tau_0) = \mathbb{E}[Y(1) - Y(0) \mid W^* = 1]$ for a well-defined subpopulation W^* and for all τ_0 **if and only if** $\mathbb{P}(a(X) \geq 0 \mid W_0 = 1) = 1$
2. When this holds, the maximum value of $\mathbb{P}(W^* = 1 \mid W_0 = 1)$ equals $a_{\max}^{-1}$, where $a_{\max}$ is the largest value of the normalized weight $a(X)$

Also provide results when τ_0 is known, when it is linear in X , and when the maximum discrepancy between two CATEs is bounded

We illustrate our results with three estimands that are often interpreted as weighted averages of treatment effects: OLS, 2SLS/IV, and TWFE

End with applications to four papers in the *AER* and *QJE*

Main tools are also implemented in the companion Stata package `causalrep`

Numerical illustration of result #2

Consider again the OLS estimand

$$\beta_{\text{OLS}} = \mathbb{E} \left[\frac{\text{var}(D \mid X)}{\mathbb{E}[\text{var}(D \mid X)]} \cdot \tau_0(X) \right]$$

and suppose the ATE is the target parameter.

The maximum value of $\mathbb{P}(W^* = 1 \mid W_0 = 1)$ for this estimand is equal to

$$a_{\max}^{-1} = \frac{\mathbb{E}[\text{var}(D \mid X)]}{\sup_{x \in \text{supp}(X)} \text{var}(D \mid X = x)}$$

Assume $X \in \{1, 2\}$ and that

$$\mathbb{P}(X = 1) = 0.2$$

$$\mathbb{P}(X = 2) = 0.8$$

$$\mathbb{P}(D = 1 \mid X = 1) = 0.4$$

$$\mathbb{P}(D = 1 \mid X = 2) = 0.1$$

$$\text{var}(D \mid X = 1) = 0.24$$

$$\text{var}(D \mid X = 2) = 0.09$$

$$\begin{aligned} \text{Maximum subpopulation size} &= (0.2 \cdot 0.24 + 0.8 \cdot 0.09) / 0.24 = 0.12 / 0.24 \\ &= 0.5 \end{aligned}$$

Numerical illustration of result #2

We can also write

$$\beta_{\text{OLS}} = 0.4 \cdot \mathbb{E}[Y(1) - Y(0) \mid X = 1] + 0.6 \cdot \mathbb{E}[Y(1) - Y(0) \mid X = 2]$$

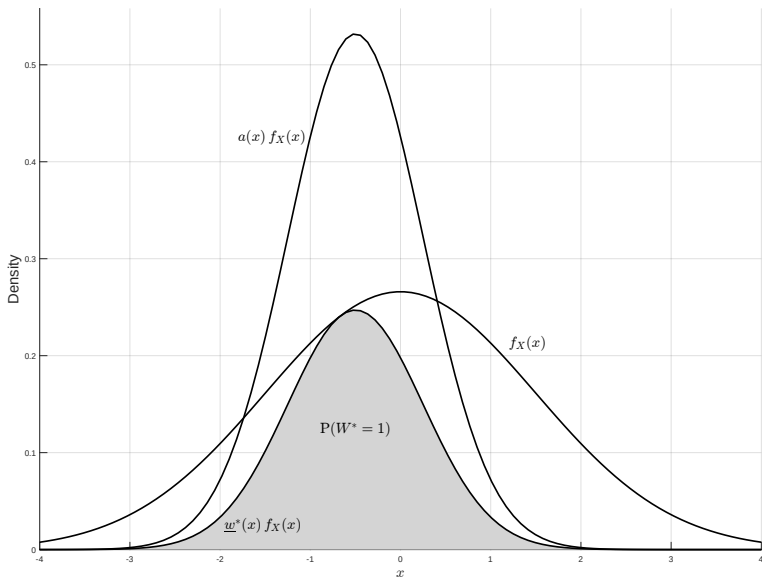
Recall that $\mathbb{P}(X = 1) = 0.2$ and $\mathbb{P}(X = 2) = 0.8$. For concreteness, assume the population has 100 units.

How do we form $\{W^* = 1\}$?

- Take all 20 units with $X = 1$.
 - They correspond to 20% of the entire population.
- Take as many units with $X = 2$ as are needed to have them correspond to 60% of the resulting subpopulation.
 - Need to take 30 units with $X = 2$.
- The size of the resulting subpopulation = $20 + 30 = 50$ units, or 50%

Cannot possibly increase the size of this subpopulation since it already includes all units with $X = 1$

Visual illustration of result #2



Notes: X is a single continuously distributed covariate.

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Related literature

OLS as weighted estimand: Angrist (1998), Aronow and Samii (2016), Słoczyński (2022), Zhao and Ding (2022), Chen (2024), Goldsmith-Pinkham, Hull, and Kolesár (2024), Humphreys (2025), Bugni, Canay, and McBride (2026)

IV/2SLS as weighted estimand: Imbens and Angrist (1994), Angrist and Imbens (1995), Kolesár (2013), Blandhol, Bonney, Mogstad, and Torgovitsky (2026), Słoczyński (2026)

TWFE as weighted estimand: de Chaisemartin and D'Haultfœuille (2020), Goodman-Bacon (2021), Sun and Abraham (2021), Athey and Imbens (2022), Borusyak, Jaravel, and Spiess (2024), Caetano and Callaway (2024), Callaway, Goodman-Bacon, and Sant'Anna (2024)

Also related to de Chaisemartin (2012)'s analysis of “external validity” of IV and Blandhol et al. (2026)'s notion of “weakly causal estimands” that guarantee sign preservation when all treatment effects have the same sign

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General setup

First question: When can a weighted estimand $\mu(a, \tau_0) = \mathbb{E}[a(X)\tau_0(X) \mid W_0 = 1]$ be written as $\mathbb{E}[Y(1) - Y(0) \mid W^* = 1]$ for some subpopulation $\{W^* = 1\}$ of $\{W_0 = 1\}$?

We assume that $\mathbb{P}(W_0 = 1) > 0$ to avoid technical issues

Recall

$$\mu(a, \tau_0) \equiv \mathbb{E}[a(X)\tau_0(X) \mid W_0 = 1]$$

We maintain the following:

Assumption (A1: Regularity)

Suppose $\mathbb{E}[\tau_0(X)^2] < \infty$, $\mathbb{E}[a(X)^2] < \infty$, and $\mathbb{E}[a(X) \mid W_0 = 1] = 1$.

Subpopulations

Let $W^* \in \{0, 1\}$ and let $\{W^* = 1\}$ denote a **subpopulation** of $\{W_0 = 1\}$ if $\{W^* = 1\} \subseteq \{W_0 = 1\}$

We impose some structure on this problem by restricting how these subpopulations may be formed

Definition (Regular subpopulations)

Let $W^* \in \{0, 1\}$ such that $\mathbb{P}(W^* = 1) > 0$. Say $\{W^* = 1\}$ is a *regular subpopulation* of $\{W_0 = 1\}$ if

1. (Inclusion) $\mathbb{P}(W_0 = 1 \mid W^* = 1) = 1$;
2. (Conditional independence) $W^* \perp\!\!\!\perp (Y(1), Y(0)) \mid X, W_0 = 1$.

Regular subpopulations

Might seem restrictive, but many natural subpopulations satisfy $W^* \perp\!\!\!\perp (Y(1), Y(0)) \mid X, W_0 = 1$. Under unconfoundedness, some examples include:

- $W^* = D$
- $W^* = \mathbb{1}(X \in \mathcal{X})$ where $\mathcal{X}$ is a subset of $\text{supp}(X)$
- $W^* = \mathbb{1}(U \leq f(X))$ where $U \perp\!\!\!\perp (X, Y(1), Y(0))$ and $U \sim \text{Unif}(0, 1)$: fraction $f(X)$ of units with covariate X

Rules out subpopulations like $W^* = \mathbb{1}(Y(1) \geq Y(0))$

Let

$$\text{SP}(W_0) = \{W^* \in \{0, 1\} : W^* \text{ is a regular subpopulation of } W_0\}.$$

Regular subpopulations: Properties

Proposition (Properties of regular subpopulations)

Let $W^* \in SP(W_0)$.

1. When $\mathbb{P}(W^* = 1 \mid W_0 = 1, X) > 0$,

$$\mathbb{E}[Y(1) - Y(0) \mid W^* = 1, X] = \mathbb{E}[Y(1) - Y(0) \mid W_0 = 1, X].$$

2. $\mathbb{E}[Y(1) - Y(0) \mid W^* = 1] = \mu(\underline{w}^* / \mathbb{E}[\underline{w}^*(X) \mid W_0 = 1], \tau_0)$, where $\underline{w}^*(x) = \mathbb{P}(W^* = 1 \mid X = x, W_0 = 1)$

Existence of causal representation

Whether we can write $\mu(a, \tau_0) = \mathbb{E}[Y(1) - Y(0) \mid W^* = 1]$ depends on what we assume about the τ_0 function

E.g., is it unrestricted? Known/fixed? In between?

To formalize this, let $\mathcal{T}$ be a class of functions from $\text{supp}(X) \rightarrow \mathbb{R}$ to which τ_0 belongs

Definition

Let

$$\mathcal{W}(a; W_0, \mathcal{T}) \equiv \{W^* \in \text{SP}(W_0) : \mu(a, \tau) = \mu(\underline{w}^* / \mathbb{E}[\underline{w}^*(X) \mid W_0 = 1], \tau) \text{ for all } \tau \in \mathcal{T}\}.$$

In words: the regular subpopulations of W_0 such that the estimand $\mu(a, \tau)$ is an average treatment effect over that subpopulation uniformly in $\tau \in \mathcal{T}$

Existence of causal representation

Definition

An estimand $\mu(a, \tau)$ has a *causal representation uniformly in $\tau \in \mathcal{T}$* if

$$\mathcal{W}(a; W_0, \mathcal{T}) \neq \emptyset.$$

Conversely there is no such representation if $\mathcal{W}(a; W_0, \mathcal{T}) = \emptyset$

Existence of causal representation: Uniformly in τ_0

Let

$$\tau_0 \in \mathcal{T}_{\text{all}} \equiv \{\tau : \mathbb{E}[\tau(X)^2] < \infty\}.$$

This case corresponds to unrestricted heterogeneity

$\mu(a, \tau_0)$ has a causal representation uniformly in $\tau_0 \in \mathcal{T}_{\text{all}}$ iff

$$\mathbb{E}[a(X)\tau_0(X) \mid W_0 = 1] = \mathbb{E}\left[\frac{\underline{w}^*(X)}{\mathbb{E}[\underline{w}^*(X) \mid W_0 = 1]}\tau_0(X) \mid W_0 = 1\right]$$

for all $\tau_0 \in \mathcal{T}_{\text{all}}$ and some $\underline{w}^*(X) \in [0, 1]$

Theorem

*Let A1 hold and let $\sup(\text{supp}(a(X) \mid W_0 = 1)) \equiv a_{\max} < \infty$. Then, there exists a causal representation uniformly in $\tau_0 \in \mathcal{T}_{\text{all}}$ **if and only if***

$$\mathbb{P}(a(X) \geq 0 \mid W_0 = 1) = 1.$$

Existence of causal representation: Uniformly in τ_0

In this case, there exists a causal representation iff weights are nonnegative

Lines up with standard intuition and criterion for weighted estimands

To see why, consider the nonnegative CATE function

$$\tau^-(X) = \mathbb{1}(a(X) \leq 0) \geq 0$$

If $\mathbb{P}(a(X) \geq 0 \mid W_0 = 1) < 1$, then

$$\mu(a, \tau^-) < 0$$

However, $\mu(a, \tau^-) < 0$ cannot equal the average effect for any subpopulation of W_0 , given that the CATE function is nonnegative

Existence of causal representation: Uniformly in τ_0

We can also construct a W^* that satisfies the assumption:

$$W^* = \mathbb{1} \left(U \leq \frac{a(X)}{a_{\max}} \right) \cdot W_0,$$

where $U \sim \text{Unif}(0, 1) \perp\!\!\!\perp (Y(1), Y(0), X, W_0)$.

Note that $W^* \in \{0, 1\}$ is a regular subpopulation of W_0 . Probability of being in W^* depends on magnitude of $a(X)$

We can show that $\mu(a, \tau_0) = \mu(\underline{w}^* / \mathbb{E}[\underline{w}^*(X) \mid W_0 = 1], \tau_0) = \mathbb{E}[Y(1) - Y(0) \mid W^* = 1]$ for all $\tau_0 \in \mathcal{T}_{\text{all}}$ where $\underline{w}^*(X) = \mathbb{P}(W^* = 1 \mid X, W_0 = 1)$

Existence of causal representation: Given τ_0

The conditions for the existence of a causal representation depend on the set of functions τ_0 belongs to

After $\mathcal{T}_{\text{all}}$, we consider the other extreme: $\mathcal{T} = \{\tau_0\}$ or τ_0 is known

Basically we want to know when there exists W^* such that $\mu(a, \tau_0) = \mathbb{E}[Y(1) - Y(0) \mid W^* = 1]$ when τ_0 is fixed

Answer: when $\mu(a, \tau_0)$ is in the convex hull of the support of $\tau_0(X)$

Existence of causal representation: Given τ_0

Theorem

*Let A1 hold. Then, there exists a causal representation for estimand $\mu \equiv \mu(a, \tau_0)$ for the given τ_0 **if and only if***

$$\mathbb{P}(\tau_0(X) \leq \mu \mid W_0 = 1) > 0 \text{ and } \mathbb{P}(\tau_0(X) \geq \mu \mid W_0 = 1) > 0.$$

Much weaker requirement than $a(X) \geq 0$ a.s.

But this result doesn't give **any guarantees** about the estimand if applied to another setting where τ_0 differs

Existence of causal representation: Intermediate case #1

We may not want to assume τ_0 is known, or that τ_0 is completely unrestricted

We briefly consider two intermediate classes. First let

$$\mathcal{T}_{\text{lin}} = \{\tau \in \mathcal{T}_{\text{all}} : \tau(X) = c + d'X, (c, d) \in \mathbb{R}^{1+d_X}\}.$$

Proposition

Suppose X has finite support and A1 holds. Then, there exists a causal representation uniformly in $\tau_0 \in \mathcal{T}_{\text{lin}}$ iff

$$\mathbb{E}[a(X)X \mid W_0 = 1] \in \text{conv}(\text{supp}(X \mid W_0 = 1)).$$

This doesn't require $a(X) \geq 0$ anymore! But requires $\tau_0(X)$ to be affine

Existence of causal representation: Intermediate case #2

For $\Delta \geq 0$, let

$$\mathcal{T}_{\text{BD}}(\Delta) = \left\{ \tau \in \mathcal{T}_{\text{all}} : \sup_{x, x' \in \text{supp}(X | W_0=1)} |\tau(x) - \tau(x')| \leq \Delta \right\}$$

Proposition

*Let A1 hold. Then, for $\Delta > 0$, there exists a causal representation uniformly in $\tau_0 \in \mathcal{T}_{\text{BD}}(\Delta)$ **iff** $\mathbb{P}(a(X) \geq 0 \mid W_0 = 1) = 1$.*

If $\Delta = 0$, there exists a causal representation uniformly in $\tau_0 \in \mathcal{T}_{\text{BD}}(0)$ for all $a(\cdot)$.

So bounding the maximum discrepancy between two CATEs using L^∞ -norm does **not** relax the requirement that $a(X) \geq 0$

Future work: additional / empirically motivated function classes

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Definitions

Suppose the researcher is interested in the target estimand $\mathbb{E}[Y(1) - Y(0) \mid W_0 = 1]$ but uses the weighted estimand $\mu(a, \tau_0)$

How to measure the validity of the weighted estimand (for this target)?

Definition (Measure of Internal Validity)

Let

$$\text{MIV}(a, W_0; \mathcal{T}) = \sup_{W^* \in \mathcal{W}(a; W_0, \mathcal{T})} \mathbb{P}(W^* = 1 \mid W_0 = 1)$$

denote the MIV of the estimand $\mu(a, \tau_0)$ over the function class $\mathcal{T}$.

The MIV is 0 if $\mathcal{W}(a; W_0, \mathcal{T})$ is empty $\iff$ there does not exist a causal representation uniformly in $\mathcal{T}$

Definitions

Can also define the following

Definition (Measure of Representativeness)

Let

$$\text{MR}(a, W_0; \mathcal{T}) = \sup_{W^* \in \mathcal{W}(a; W_0, \mathcal{T})} \mathbb{P}(W^* = 1)$$

denote the MR of the estimand $\mu(a, \tau_0)$ over the function class $\mathcal{T}$.

The MR is effectively the MIV of $\mu(a, \tau_0)$ with respect to the ATE = $\mathbb{E}[Y(1) - Y(0)]$

Motivation

Why should we care about these measures?

1. We have shown that common criteria for “reasonable estimands” are necessary and sufficient for the existence of W^* . So, applied researchers implicitly care about W^* , and it seems useful to understand it better.

Motivation (continued)

Why should we care about these measures?

2. We argue that the MIV and MR are valuable diagnostics for applied research—it is good when they are large!
 - Mogstad and Torgovitsky (2024): “Target parameters that reflect larger subpopulations of the population of interest are more interesting than those that reflect smaller and more specific subpopulations.”
 - Similarly, in a setting with multiple instrumental variables, van 't Hoff, Lewbel, and Mellace (2025) argue that the largest subpopulation of compliers is generally more interesting than other complier subpopulations.

Motivation (continued)

Why should we care about these measures?

3. As we will see, $\mu(a, \tau_0)$ coupled with $\mathbb{P}(W^* = 1 \mid W_0 = 1)$ and a support restriction allows us to bound $\mathbb{E}[Y(1) - Y(0) \mid W_0 = 1]$.
 - These bounds are really simple! Don't require estimating anything beyond our measure, assuming that $\mu(a, \tau_0)$ has been estimated.

Quantifying internal validity: Uniformly in τ_0

Theorem (Uniform internal validity)

Let A1 hold and let $\sup(\text{supp}(a(X) \mid W_0 = 1)) \equiv a_{\max} < \infty$.

If $\mathbb{P}(a(X) \geq 0 \mid W_0 = 1) = 1$, then

$$\text{MIV}(a, W_0; \mathcal{T}_{\text{all}}) = a_{\max}^{-1}.$$

Intuition

Intuition for this formula can be obtained from

$$\begin{aligned}\mu(a, \tau_0) &= \mathbb{E}[a(X)\tau_0(X) \mid W_0 = 1] \\ &= \mathbb{E}\left[\frac{\underline{w}^*(X)}{\mathbb{E}[\underline{w}^*(X) \mid W_0 = 1]} \tau_0(X) \mid W_0 = 1\right]\end{aligned}$$

which must hold for all $\tau_0 \in \mathcal{T}_{\text{all}}$ for some $\underline{w}^*(X) = \mathbb{P}(W^* = 1 \mid X, W_0 = 1)$

Intuition (continued)

This implies $a(X)$ is proportional to $\underline{w}^*(X) = \mathbb{P}(W^* = 1 \mid X, W_0 = 1)$

Note that $\underline{w}^*(X) \in [0, 1]$ and $\mathbb{P}(W^* = 1 \mid W_0 = 1) = \mathbb{E}[\underline{w}^*(X) \mid W_0 = 1]$

So we want to maximize $\mathbb{E}[\underline{w}^*(X) \mid W_0 = 1]$ over $\underline{w}^*$ subject to $\underline{w}^*(X) \in [0, 1]$ and $\underline{w}^*(X)$ prop. to $a(X)$

Solution: select the **largest** multiple of $a(X)$ that is $\in [0, 1]$ w.p. 1 given $W_0 = 1$, which is given by

$$\underline{w}^*(X) = \frac{a(X)}{a_{\max}}.$$

Intuition (continued)

This corresponds to a regular subpopulation:

$$W^* = \mathbb{1} \left(U \leq \frac{a(X)}{a_{\max}} \right) \cdot W_0$$

where $U \sim \text{Unif}(0, 1)$ is independent from other variables

Note that $\{W^* = 1\}$ is a subset of $\{W_0 = 1\}$ by construction, and $\mathbb{P}(W^* = 1 \mid W_0 = 1) = \mathbb{E}[a(X)/a_{\max} \mid W_0 = 1] = a_{\max}^{-1}$

Diagnostics based on $a_{\max}^{-1}$

Recall that $a_{\max}^{-1}$ is the MIV under unrestricted heterogeneity when the weights are nonnegative

Regardless of the sign of the weights, we can decompose the target estimand as

$$\begin{aligned}\mathbb{E}[Y(1) - Y(0) \mid W_0 = 1] \\&= \mathbb{E}[\tau_0(X) \mid W_0 = 1] \\&= \mathbb{E}[(a(X)a_{\max}^{-1} + 1 - a(X)a_{\max}^{-1}) \tau_0(X) \mid W_0 = 1] \\&= \mu(a, \tau_0)a_{\max}^{-1} + \mathbb{E}[(1 - a(X)a_{\max}^{-1}) \tau_0(X) \mid W_0 = 1].\end{aligned}$$

The first term depends only on $\mu(a, \tau_0)$ and $a_{\max}^{-1}$

Since $1 - a(X)a_{\max}^{-1} \geq 0$ a.s. given $W_0 = 1$, restrictions on $\tau_0(X)$ can be used to bound the second term, even when some weights are negative

Bounds under a support restriction

Suppose $\mu(a, \tau_0)$ and $a_{\max}^{-1}$ are known and

$$\mathbb{P}(B_\ell \leq \tau_0(X) \leq B_u \mid W_0 = 1) = 1,$$

where $[B_\ell, B_u]$ is a known interval containing $\mu(a, \tau_0)$

Then the following bounds are sharp:

$$\mathbb{E}[Y(1) - Y(0) \mid W_0 = 1] \in \left[\mu(a, \tau_0) a_{\max}^{-1} + B_\ell (1 - a_{\max}^{-1}), \right. \\ \left. \mu(a, \tau_0) a_{\max}^{-1} + B_u (1 - a_{\max}^{-1}) \right].$$

The width of the bounds is

$$(1 - a_{\max}^{-1})(B_u - B_\ell)$$

No sign restriction on $a(X)$ is required

Further sharp bounds

Variance restriction. Suppose $\mathbb{P}(a(X) \geq 0 \mid W_0 = 1) = 1$ and

$$\text{var}(\tau_0(X) \mid W_0 = 1) \leq \sigma_\tau^2.$$

Then

$$\mathbb{E}[Y(1) - Y(0) \mid W_0 = 1] \in \left[\mu(a, \tau_0) - \sqrt{a_{\max} - 1} \sigma_\tau, \right. \\ \left. \mu(a, \tau_0) + \sqrt{a_{\max} - 1} \sigma_\tau \right].$$

Fraction of negative weights. Without any sign restriction,

$$\mathbb{P}(a(X) < 0 \mid W_0 = 1) \leq 1 - a_{\max}^{-1}.$$

For example, if $a_{\max}^{-1} = 0.95$, at most 5% of the weights can be negative

Covariates in the implicit subpopulation

Suppose $\mathbb{P}(a(X) \geq 0 \mid W_0 = 1) = 1$, and let W^* denote a largest implicit subpopulation uniformly in τ_0

For any integrable function g , its average in W^* is identified:

$$\mathbb{E}[g(X) \mid W^* = 1] = \mathbb{E}[a(X)g(X) \mid W_0 = 1].$$

Thus, we can recover

- mean covariate values by setting $g(X) = X$;
- the covariate distribution by setting $g(X) = \mathbb{1}(X \leq x)$.

Comparing these quantities with their counterparts in W_0 describes which units the estimand overrepresents or underrepresents

Quantifying internal validity: Given τ_0

We now “fix” τ_0 . Consider

$$\text{MIV}(a, W_0; \{\tau_0\}) = \sup_{W^* \in \mathcal{W}(a; W_0, \{\tau_0\})} \mathbb{P}(W^* = 1 \mid W_0 = 1),$$

the MIV of $\mu(a, \tau_0)$ for the given CATE function

This can be written as an infinite-dimensional linear programming problem:

$$\begin{aligned} \max_{\underline{w}^*} \mathbb{E}[\underline{w}^*(X) \mid W_0 = 1] \quad \text{s.t.} \quad & \mathbb{E}[(\tau_0(X) - \mu(a, \tau_0))\underline{w}^*(X) \mid W_0 = 1] = 0 \\ & \underline{w}^*(X) \in [0, 1] \end{aligned}$$

If X has finite discrete support: can easily be solved numerically

If X has continuous components: much harder!

But, there is an analytical solution

Quantifying internal validity: Given τ_0

Theorem (Internal validity given τ_0)

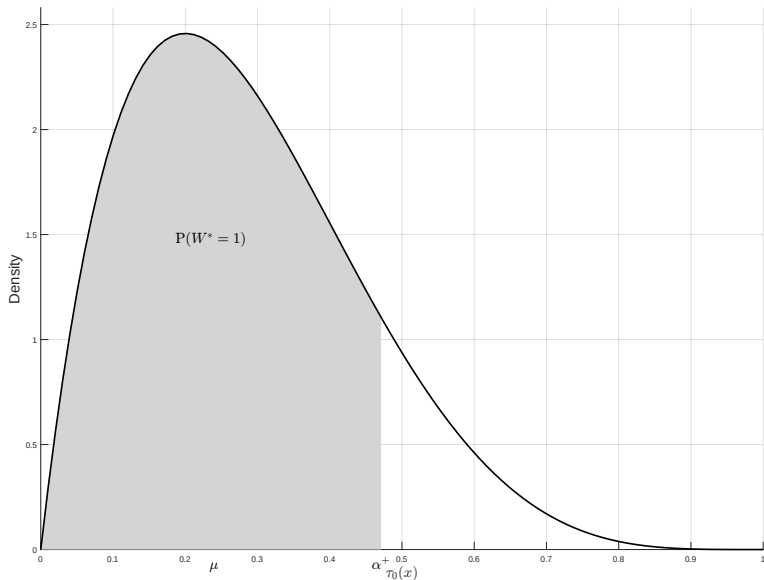
Let A1 hold. Let $\mu_0 = \mu(a, \tau_0)$, $T_\mu = \tau_0(X) - \mu_0$, and $E_0 = \mathbb{E}[Y(1) - Y(0) \mid W_0 = 1]$. If $\mu_0 \in \mathcal{S}(\tau_0; W_0) \equiv \{t \in \mathbb{R} : \mathbb{P}(\tau_0(X) \leq t \mid W_0 = 1) > 0 \text{ and } \mathbb{P}(\tau_0(X) \geq t \mid W_0 = 1) > 0\}$,

$$\text{MIV}(a, W_0; \{\tau_0\}) = \begin{cases} \mathbb{P}(T_\mu \leq \alpha^+ \mid W_0 = 1) - \frac{\mathbb{E}[T_\mu \mathbb{1}(T_\mu \leq \alpha^+) \mid W_0 = 1]}{\alpha^+} \mathbb{1}(\alpha^+ > 0) & \text{if } \mu_0 < E_0 \\ \mathbb{P}(T_\mu \geq \alpha^- \mid W_0 = 1) - \frac{\mathbb{E}[T_\mu \mathbb{1}(T_\mu \geq \alpha^-) \mid W_0 = 1]}{\alpha^-} \mathbb{1}(\alpha^- < 0) & \text{if } \mu_0 > E_0 \\ 1 & \text{if } \mu_0 = E_0, \end{cases}$$

where $\alpha^+ = \inf\{\alpha \geq 0 : \mathbb{E}[T_\mu \mathbb{1}(T_\mu \leq \alpha) \mid W_0 = 1] \geq 0\}$ and $\alpha^- = \sup\{\alpha \leq 0 : \mathbb{E}[T_\mu \mathbb{1}(T_\mu \geq \alpha) \mid W_0 = 1] \leq 0\}$.

If $\mu_0 \notin \mathcal{S}(\tau_0; W_0)$, then $\text{MIV}(a, W_0; \{\tau_0\}) = 0$.

Visual illustration of theorem



Notes: The figure assumes that $\tau_0(X)$ is continuously distributed, that $W_0 = 1$ almost surely, and that $\mu < \mathbb{E}[\tau_0(X)] = \text{ATE}$.

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Three examples

The paper studies these three theoretical examples in detail:

1. OLS under unconfoundedness
2. IV/2SLS with covariates
3. TWFE with variation in treatment timing

Example #1: OLS

Let unconfoundedness and overlap hold:

$$(Y(1), Y(0)) \perp\!\!\!\perp D \mid X \quad \text{and} \quad \mathbb{P}(D = 1 \mid X) \in (0, 1) \text{ a.s.}$$

Let $p(X) \equiv \mathbb{P}(D = 1 \mid X)$ be linear in X (saturated specification or “rich covariates”)

Let β_{OLS} be the coefficient on D in a regression of Y on $(1, D, X)$

Proposition

Under the above assumptions,

$$\beta_{OLS} = \mathbb{E} \left[\frac{\text{var}(D \mid X)}{\mathbb{E}[\text{var}(D \mid X)]} \cdot \mathbb{E}[Y(1) - Y(0) \mid X] \right],$$

where $\text{var}(D \mid X) = p(X)(1 - p(X))$

Example #1: OLS

These representations are not always unique: compare

$$\mu(a, \tau_0) \equiv \mathbb{E}[a(X)\tau_0(X) \mid W_0 = 1]$$

and

$$\begin{aligned}\beta_{\text{OLS}} &= \mathbb{E}\left[\frac{p(X)(1-p(X))}{\mathbb{E}[p(X)(1-p(X))]} \cdot \mathbb{E}[Y(1) - Y(0) \mid X]\right] \\ &= \mathbb{E}\left[\frac{1-p(X)}{\mathbb{E}[1-p(X) \mid D=1]} \cdot \mathbb{E}[Y(1) - Y(0) \mid D=1, X] \mid D=1\right]\end{aligned}$$

Now $W_0 = D$ and $a(X) = (1-p(X))/\mathbb{E}[1-p(X) \mid D=1]$

Internal validity of OLS

The OLS estimand is representative of this fraction of the population:

$$\text{MIV}(a_{\text{OLS}}, 1; \mathcal{T}_{\text{all}}) = \frac{\mathbb{E}[p(X)(1 - p(X))]}{\sup_{x \in \text{supp}(X)} p(x)(1 - p(x))}.$$

When $1/2 \in \text{supp}(p(X))$, this equals

$$\text{MIV}(a_{\text{OLS}}, 1; \mathcal{T}_{\text{all}}) = 4\mathbb{E}[p(X)(1 - p(X))] \leq 4\text{var}(D)$$

so we can easily obtain a crude bound for it

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Its internal validity **for the ATT** is given by

$$\text{MIV}(a_{\text{OLS},2}, D; \mathcal{T}_{\text{all}}) = \frac{\mathbb{E}[p(X)(1 - p(X))]}{\mathbb{P}(D = 1) \cdot \sup_x (1 - p(x))} \leq \frac{\mathbb{P}(D = 0)}{1 - \inf_x p(x)}$$

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These results are consistent with the criticism in Słoczyński (2022) without relying on some of the parametric restrictions in that paper.

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Estimation

When $\mathcal{T} = \mathcal{T}_{\text{all}}$ and the weights are nonnegative, a sample analog estimator of the MIV is

$$\hat{a}_{\max}^{-1} \equiv \frac{\frac{1}{n} \sum_{i=1}^n \hat{\alpha}(X_i) \hat{w}_0(X_i)}{\frac{1}{n} \sum_{i=1}^n \hat{w}_0(X_i) \cdot \max_{i: \hat{w}_0(X_i) > c_n} \hat{\alpha}(X_i)},$$

where $\alpha(X)$ is the non-normalized weight function

Technical issues:

1. Uniform convergence of $\hat{\alpha}(x)$ to $\alpha(x)$? Depends on nature of $\text{supp}(X)$
2. Non-differentiability of the sup/max operator $\Rightarrow$ standard bootstrap(s) invalid
3. We provide confidence intervals based on the nonstandard bootstrap of Fang and Santos (2019)

Empirical applications

In the paper, we apply the proposed tools to four *AER* and *QJE* articles, which study the causal effects of:

- cash transfers using OLS (Aizer, Eli, Ferrie, and Lleras-Muney 2016)
- well failure using OLS (Blakeslee, Fishman, and Srinivasan 2020)
- health insurance using IV (Finkelstein et al. 2012)
- unilateral divorce laws using DID/TWFE (Stevenson and Wolfers 2006)

Causal effects of well failure

Effects of borewell failure on rural households in India (Blakeslee, Fishman, and Srinivasan 2020)

The data consist of 891 households in the Indian state of Karnataka

The timing of borewell failure is assumed to be as good as randomly assigned conditional on borewell age and location

The treatment is whether the household's first borewell had failed by the survey date

The authors estimate effects on 50 outcomes using two common specifications. Specification #2 adds drilling-year fixed effects to the village indicators and other covariates in specification #1

Results

Internal validity of OLS for the effects of well failure

	Specification #1		Specification #2	
	ATE	ATT	ATE	ATT
$\hat{a}_{\max}^{-1}$	0.7855	0.3600	0.7251	0.2628
$\hat{\mathbb{P}}(a(X) < 0 \mid W_0 = 1)$	0.0088	0.0143	0.0413	0.0533
$\widehat{\text{MIV}}$ (uniformly in τ_0)	0	0	0	0

Causal effects of health insurance

Effects of Medicaid enrollment on health and economic outcomes (Finkelstein et al. 2012)

In 2008, Oregon randomly selected eligible low-income adults for the opportunity to apply for Medicaid. Only some selected individuals enrolled

Our sample consists of 23,741 survey respondents. The treatment is Medicaid enrollment, and randomized selection is an instrument conditional on household size and survey wave

The original paper uses a noninteracted IV specification for 30 outcomes. We also consider saturated 2SLS with multiple interacted instruments

Results

Internal validity of IV and 2SLS for the effects of health insurance

	β_{IV}	β_{2SLS}
$\hat{a}_{\max}^{-1}$	0.9413	0.7460
$\hat{\mathbb{P}}(a(X) < 0 \mid W_0 = 1)$	0	0
$\widehat{MIV}$ (uniformly in τ_0)	0.9413	0.7460
$\widehat{MR}$ (uniformly in τ_0)	0.2720	0.2156

Causal effects of unilateral divorce laws

Effects of unilateral divorce laws in the U.S. on female suicide (Stevenson and Wolfers 2006)

Between 1969 and 1985, 37 states (including D.C.) enabled each spouse to seek divorce without the other spouse's consent

Our sample consists of 41 states observed over the 1964–1996 period. We removed “always-treated” states

The outcome of interest is the state- and year-specific female suicide rate (per million women)

The treatment is whether the state allowed unilateral divorce in a given year

Results

Internal validity of TWFE for the effects of unilateral divorce

A. Estimates of the effects of unilateral divorce laws		
TWFE	ATT	
	Callaway & Sant'Anna	Wooldridge
-0.604	-10.220	-5.530
(2.622)	(3.086)	(3.650)

B. Internal validity of TWFE based on dCDH weights

$\widehat{a}_{\max}^{-1}$	0.1180	
$\widehat{\mathbb{P}}(a(X) < 0 \mid W_0 = 1)$	0.2456	
	uniformly in τ_0	given τ_0
$\widehat{\text{MIV}}$	0	0.6216
$\widehat{\text{MR}}$	0	0.3873

Results

Internal validity of TWFE for the effects of unilateral divorce

A. Estimates of the effects of unilateral divorce laws

TWFE	ATT	
	Callaway & Sant'Anna	Wooldridge
-0.604 (2.622)	-10.220 (3.086)	-5.530 (3.650)

C. Internal validity of TWFE based on G-B weights

$\widehat{a}_{\max}^{-1}$	0.2246	
$\widehat{\mathbb{P}}(a(X) < 0 \mid W_0 = 1)$	0	
	uniformly in τ_0	given τ_0
$\widehat{\text{MIV}}$	0.2246	0.7707
$\widehat{\text{MR}}$	0.1400	0.4802

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Conclusion

In this paper, we studied the representativeness and internal validity of a class of weighted estimands, which includes the popular OLS, IV/2SLS, and TWFE estimands

We examined the conditions under which such estimands can be written as the average treatment effect for some (possibly latent) subpopulation

Derived the sharp upper bound on the size of that subpopulation

Valuable diagnostic for empirical research: can be used to compare different estimands or give simple bounds for target estimands

Main tools are also implemented in the companion Stata package `causalrep`

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Thank you!